AI-BASED VOICE CLONING AND TEXT-TO-SPEECH SYSTEM WITH AI-GENERATED VOICE DETECTION USING PYTHON AND MATLAB

GROUP 7

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1. Abstract

- The project combines two functions: (1) text-to-speech/voice cloning and (2) detection of AI-generated speech.
- Python is used for AI, deep learning and audio processing.
- MATLAB is used for speech-signal analysis and visualization.
- The system is planned with free/open-source software and pretrained models where possible; no paid API is required.
- The detector classifies an input recording as Human Voice or AI-Generated Voice.

2. Main Objectives

- Convert user-entered text into speech.
- Generate speech using an authorized reference voice with a free/open-source pretrained TTS/voice-cloning model.
- Preprocess voice recordings using Python.
- Generate mel-spectrograms from audio.
- Train a CNN classifier to distinguish human and AI-generated speech.
- Analyze original and generated audio in MATLAB.
- Display a simple prediction: Human Voice / AI-Generated Voice with confidence.
- Evaluate the detector using accuracy, precision, recall, F1-score and a confusion matrix.

3. Proposed System

Stage	What We Will Build
1. Voice Input	Record or select a permitted voice sample in WAV format.

2. Preprocessing	Convert to a common sampling rate, normalize audio and trim long silent portions.
3. Voice Cloning / TTS	Use a free/open-source pretrained model to generate speech from text and an authorized reference voice.
4. Audio Dataset	Use genuine speech samples plus AI-generated samples for two-class training.
5. Feature Representation	Convert speech into mel-spectrograms.
6. CNN Detector	Train a CNN to classify Human vs AI-Generated speech.
7. MATLAB Analysis	Plot waveform, spectrogram, MFCC and pitch for selected samples.
8. Final Result	Show predicted class and confidence; compare results with known labels during testing.

4. Voice Cloning / TTS Module

- Input: text entered by the user + an authorized reference voice.
- The reference voice is used to capture speaker characteristics through the selected pretrained model.
- The TTS model generates speech and saves the result as a WAV file.
- The generated file is used for playback, testing and MATLAB analysis.
- A pretrained model is used instead of training a large TTS model from scratch.
- Only voices for which permission is available will be used in the demonstration.

5. AI-Generated Voice Detection

- Two classes: Human and AI-Generated.
- Human samples come from a suitable free speech dataset and/or permitted recordings.
- AI samples are generated using the selected TTS/voice-cloning system.
- Audio is converted into mel-spectrograms.
- A CNN learns patterns from the spectrograms.
- The output gives probabilities for both classes.
- Example: Human: 12% | AI-Generated: 88%.

6. Proposed CNN Structure

Layer	Purpose
Input	Mel-spectrogram image/tensor.
Convolution + ReLU	Learn local time-frequency patterns.
Max Pooling	Reduce feature-map size.
Convolution + ReLU	Learn higher-level acoustic patterns.
Max Pooling	Further reduce dimensions.
Flatten / Global Pooling	Prepare learned features for classification.
Fully Connected	Combine learned features.

Output	Human / AI-Generated probability.

7. MATLAB Module

- Waveform: amplitude versus time.
- Spectrogram: frequency content versus time.
- MFCC: commonly used speech features.
- Pitch: fundamental-frequency variation.
- Comparison: compare selected human and AI-generated samples.
- MATLAB is used mainly for signal analysis and visualization; CNN training and inference are performed in Python.

8. Free Technology Stack

Component	Tool / Approach
Programming	Python
Deep Learning	PyTorch
Audio Processing	Librosa / torchaudio / SoundFile
TTS / Voice Cloning	Free/open-source pretrained model selected during implementation
Detection Model	CNN using PyTorch
Evaluation	scikit-learn
Signal Analysis	MATLAB
Dataset	Free/open speech dataset + locally generated AI samples
Interface	Simple Python program or free local GUI, if required

9. Dataset and Training Plan

- Use human speech recordings with suitable permission/licensing.
- Generate AI speech using the selected TTS system.
- Keep the two classes reasonably balanced.
- Split data into training, validation and test sets.
- Prefer speaker-independent testing so the CNN is not simply memorizing speakers.
- Train on the training set and use validation data to choose model settings.
- Evaluate on samples that were not used for training.

10. Evaluation Metrics

Metric	Purpose
Accuracy	Overall percentage of correct predictions.
Precision	How often predicted AI samples are actually AI-generated.
Recall	How many AI-generated samples are detected.
F1-score	Balance between precision and recall.
Confusion Matrix	Shows correct and incorrect predictions for both classes.

11. Expected Output

- Text input → generated speech WAV file.
- Reference voice → synthesized speech resembling the authorized speaker.

- Audio input → mel-spectrogram → CNN → Human/AI prediction.
- MATLAB displays waveform, spectrogram, MFCC and pitch.
- Evaluation report shows accuracy, precision, recall, F1-score and confusion matrix.

12. Applications

- Personalized text-to-speech.
- Educational audio and small-scale audiobook generation.
- Accessibility support.
- Demonstration of voice restoration concepts using an authorized voice.
- Voice-based media and content experiments.
- Basic screening of potentially AI-generated audio.

13. Limitations

- The detector will not be guaranteed to detect every type of AI-generated speech.
- Performance may decrease on generators or audio conditions not represented in training.
- A small dataset may limit generalization.
- Voice-cloning quality depends on the selected pretrained model and reference recording.
- A GPU may improve training speed, but the project should be designed around available hardware.
- The project is an academic prototype, not a professional forensic authentication system.

14. Ethical and Responsible Use

- Use voice samples only with the speaker's permission.
- Do not use the system to impersonate people or create deceptive recordings.
- Clearly identify generated audio during demonstrations.
- Treat detector output as a prediction, not absolute proof.
- Protect project voice recordings and avoid unnecessary storage.

15. Future Scope

- Increase dataset size and speaker diversity.
- Test the detector on additional free/open-source TTS systems.
- Improve robustness against noise and compression.
- Compare CNN with another lightweight detection model.
- Develop a simple local GUI for synthesis and detection.

16. Final Project Flow

Reference Voice → Preprocessing → TTS/Voice Cloning → Generated WAV → MATLAB Analysis

Human/Generated WAV → Preprocessing → Mel-Spectrogram → CNN → Human / AI-Generated → Confidence Score